



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VELMURUGA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

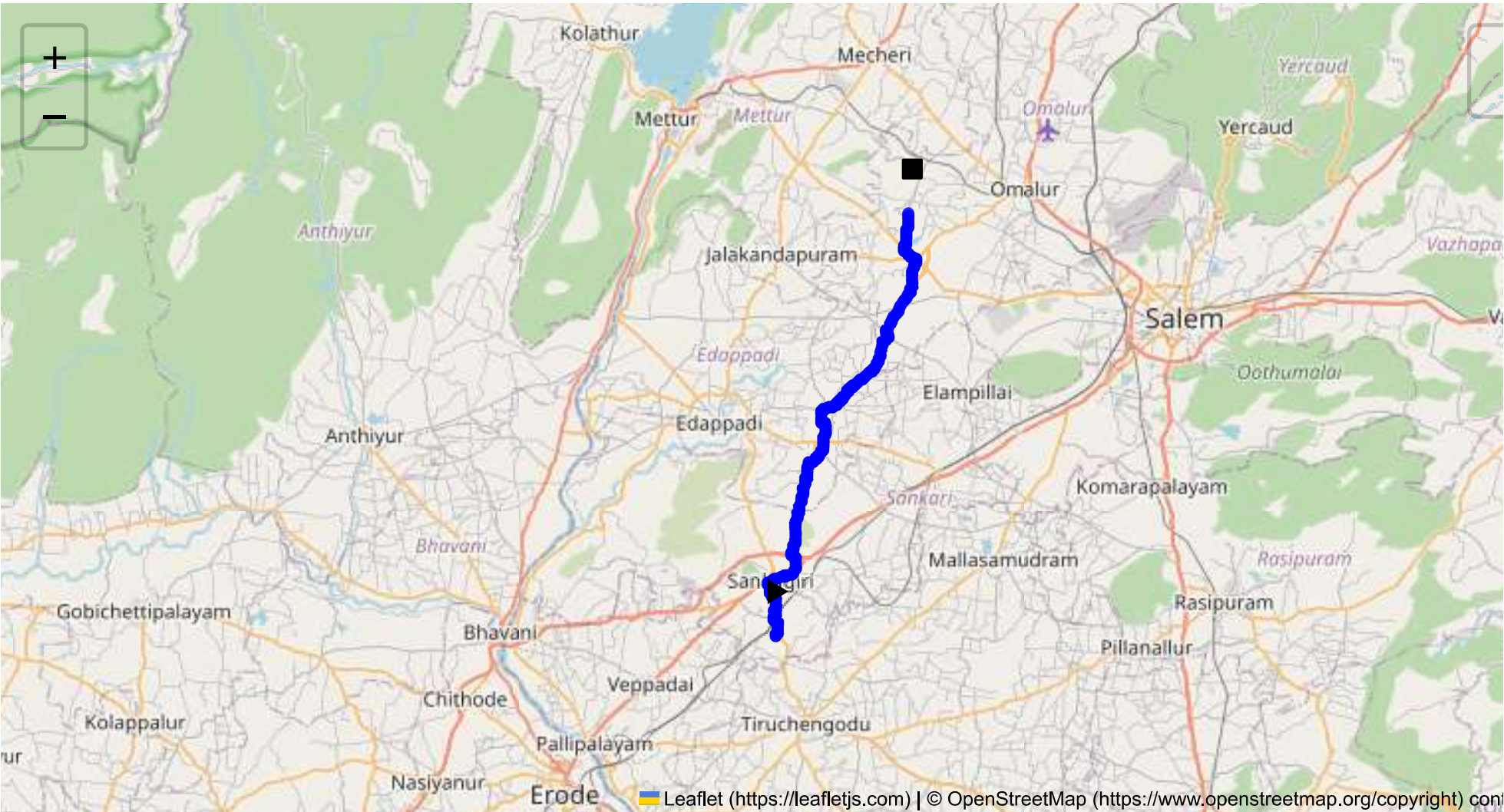
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 38.29 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.726705, 77.966782)



Welcome to the Journey Risk Management Study

Analyzing the route from CVQF+23W, Sangagiri to PXG8+MP, Annamakkarai, Tamil Nadu, India requires careful consideration of multiple factors to ensure a comprehensive safety assessment for truck drivers carrying hazardous materials.

1. Overview of the Route Map:

- The primary route spans approximately 38.29 kilometers through rural Tamil Nadu, primarily using state highways. The path is relatively straightforward, passing through small towns and villages.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- The route lies in a tropical climate zone, experiencing hot and humid conditions. Monsoon rains (June to September) can lead to slippery roads and reduced visibility. Occasional thunderstorms or cyclones can occur, posing additional hazards.

3. Analysis of Traffic Patterns:

- Traffic is generally moderate but can increase at peak commuting hours, typically 8-10 AM and 5-7 PM. Congestion may occur near market towns and urban areas along the route, including near Sangagiri.

4. Assessment of Road Quality and Infrastructure:

- The roads are generally in fair condition, with some segments possibly affected by potholes or narrower rural paths. Infrastructure such as bridges is generally sound but may not handle excessive loads efficiently.

5. Suggestions for Alternative Routes:

- In emergencies, an alternative route may include diverting through SH79 and NH44, though it might extend the travel distance and time.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Regulations include restrictions on timing (avoiding night travel), specific routes allowed for hazardous materials, and weight limits. Adherence to Tamil Nadu State transport guidelines is mandatory.

7. Overview of Historical Incidents:

- While specific historical data for this route may be limited, incidents such as overturned vehicles or hazardous material spills are risks in rural roads due to high speeds and limited maneuverability.

8. Environmental Considerations and Sensitive Areas:

- The route may pass near agricultural lands and small water bodies. Care must be taken to prevent contamination from spills. Areas near schools or markets require heightened caution.

9. Analysis of Communication Coverage:

- Communication signals are typically strong near urban centers but may have weaker coverage or dead zones in remote rural stretches.

10. Estimated Emergency Response Times:

- Emergency response times are quicker near villages and towns, averaging 20-30 minutes. Remote areas could experience delays, potentially exceeding 45 minutes.

11. Overall Summary of Risk Assessment:

- The route presents moderate risk primarily due to weather conditions, traffic variations, and infrastructure constraints. Key risks include vehicular accidents, delays due to narrow roads, and

potential for adverse weather. Drivers should be well-trained in handling hazardous materials, aware of local regulations, and prepared for alternative routes. Preemptive measures, including real-time monitoring of weather, traffic conditions, and frequent communication checks, are essential for safe transit.

This comprehensive analysis will help ensure the safe transportation of hazardous materials along this specific route in Tamil Nadu.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	Medium	11.48363, 77.88780	30 KM/Hr
7	Turn	Medium	11.49451, 77.88489	30 KM/Hr
8	Turn	High	11.55614, 77.89764	15 KM/Hr
9	Turn	High	11.58317, 77.90647	15 KM/Hr
10	Turn	Medium	11.59061, 77.90583	30 KM/Hr
11	Turn	Medium	11.63840, 77.94820	30 KM/Hr
12	Turn	Medium	11.69436, 77.97102	30 KM/Hr
13	Blind Spot	Blind Spot	11.69590, 77.97114	10 KM/Hr
14	Turn	High	11.70129, 77.96360	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
11	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
12	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
13	hospital	Veda Maruthuvamanai Hospital	11.6956155, 77.9678122	30 km/h	Medium
14	clinic	E.V.P Vairavel Clinic	11.6954975, 77.9667903	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48363, 77.88780



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49451, 77.88489



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.55614, 77.89764



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58317, 77.90647



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59061, 77.90583



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63840, 77.94820



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69436, 77.97102



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69590, 77.97114



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70129, 77.96360